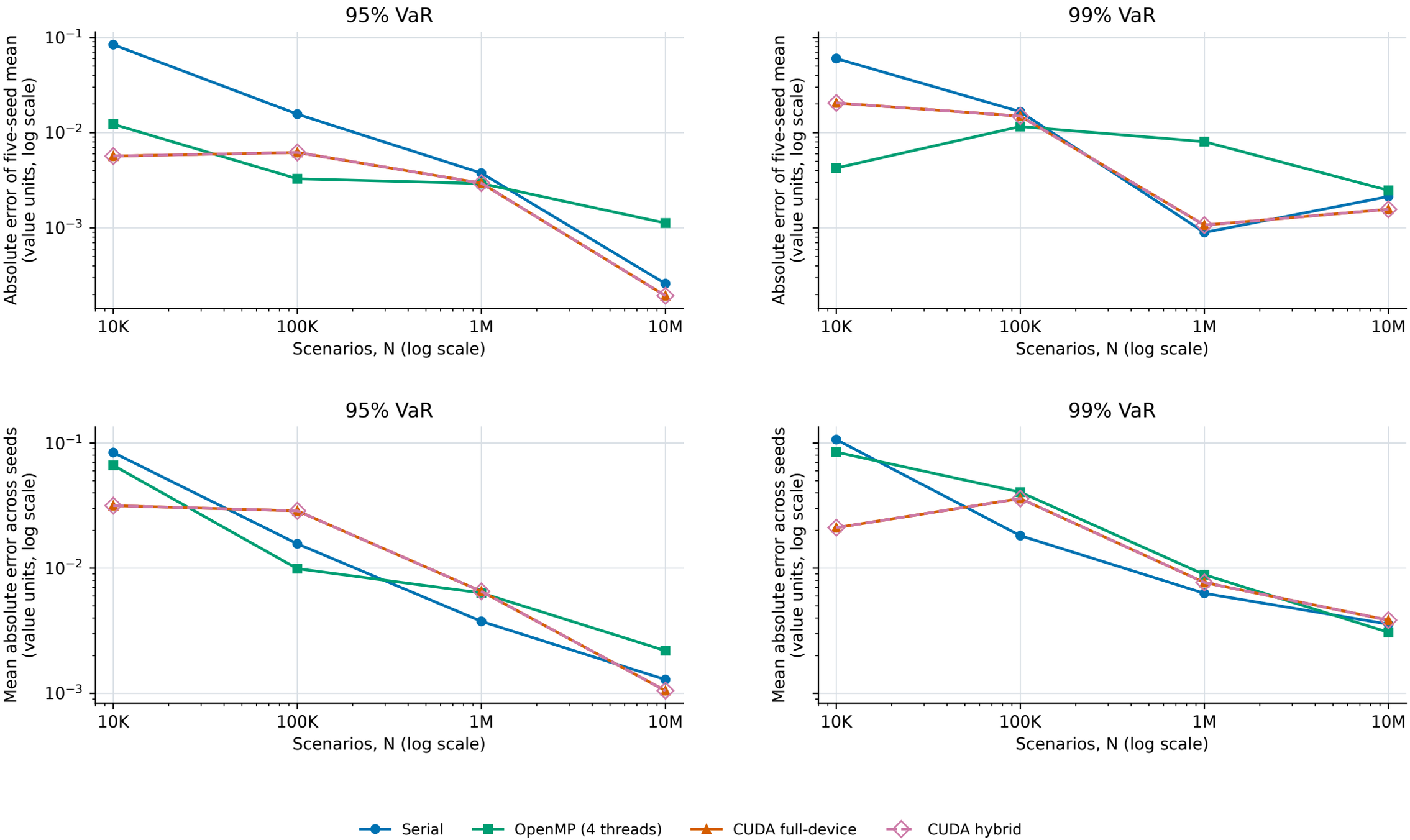
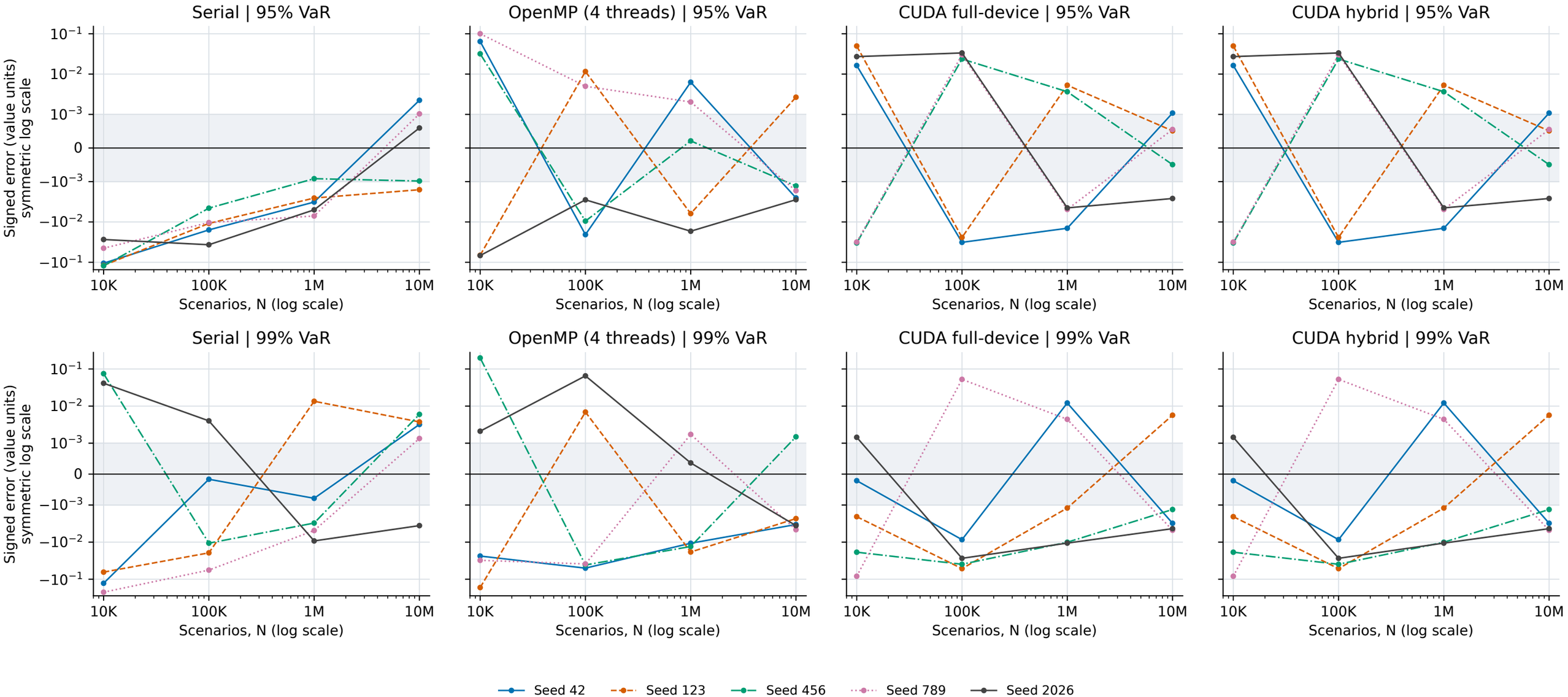
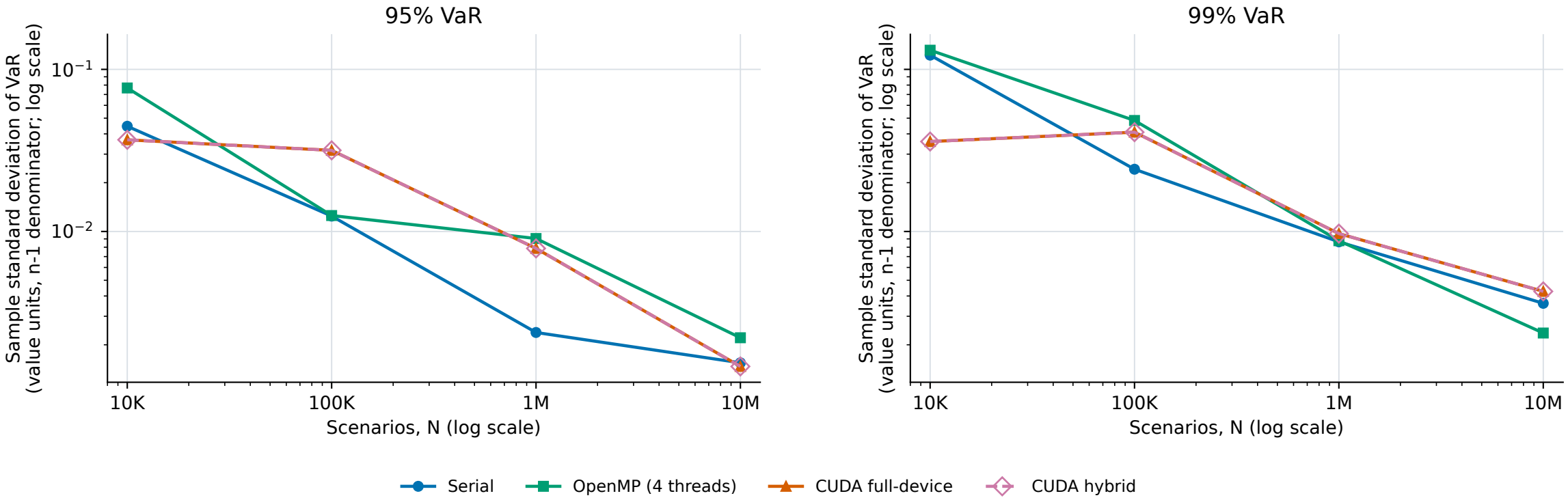
Analytical error versus scenario count



One synthetic fixture; five fixed seeds. Descriptive errors do not replace 10M acceptance.
Full-device and hybrid values overlap exactly in this dataset; no equivalence guarantee is implied.

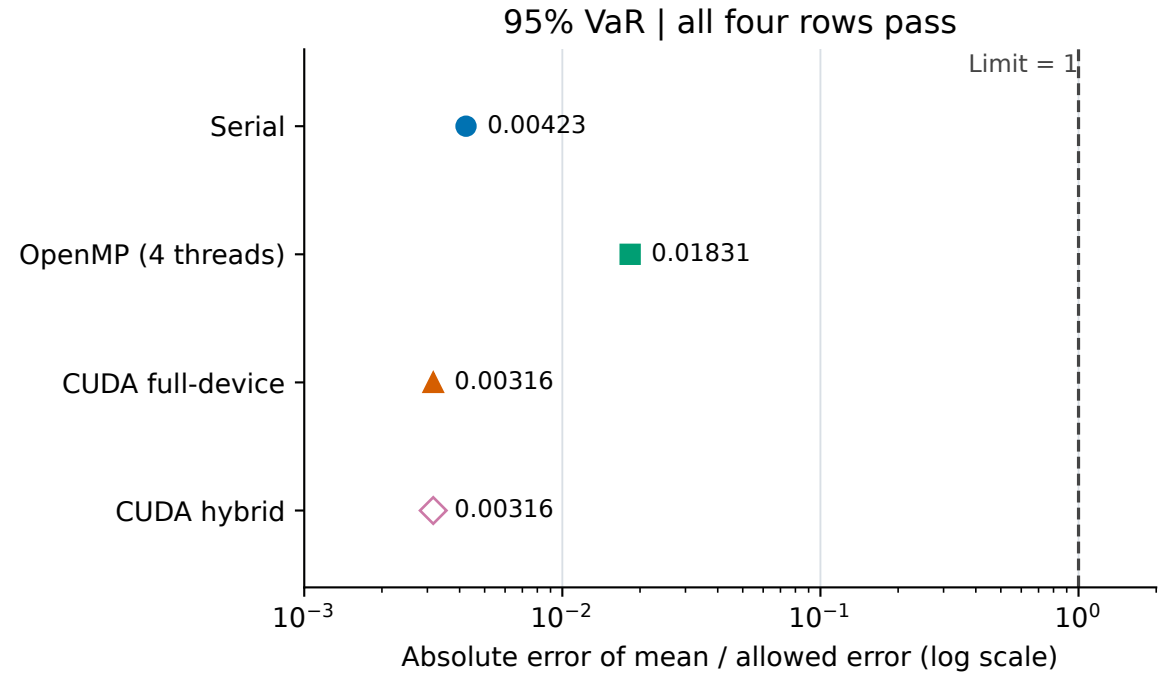

Seed-to-seed signed analytical errors
Serial | 95% VaR
OpenMP (4 threads) | 95% VaR
CUDA full-device | 95% VaR
CUDA hybrid | 95% VaR
Serial | 99% VaR
OpenMP (4 threads) | 99% VaR
CUDA full-device | 99% VaR
CUDA hybrid | 99% VaR
Signed error (value units)
symmetric log scale
10^{-1}
10^{-2}
10^{-3}
0
-10^{-3}
-10^{-2}
-10^{-1}
10K
100K
1M
10M
Scenarios, N (log scale)
Seed 42
Seed 123
Seed 456
Seed 789
Seed 2026
Gray band: linear y-axis region |error| <= 0.001; raw signs and zeros are preserved.
Lines identify fixed seeds across N; same seed labels are not common inputs across engines.

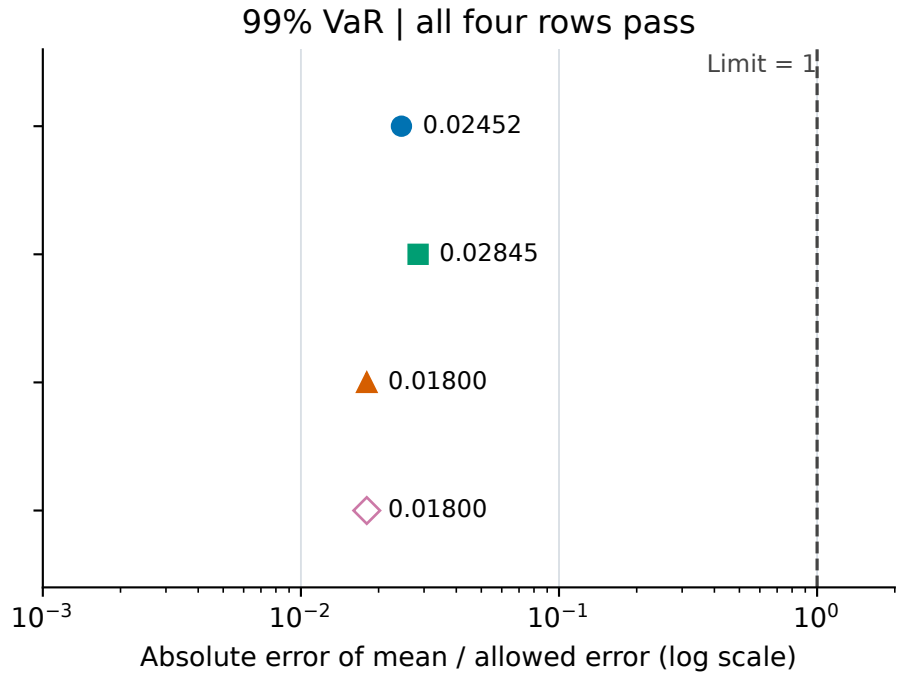
Variation across the five seeds



Sample spread is not a confidence interval or universal error guarantee.
Overlapping GPU curves retain their original values.

Separate analytical acceptance at N = 10,000,000





Acceptance: |mean(VaR) - reference| <= 0.01 max(|reference|, 0.01 V0).
These eight rows reuse existing 10M observations; individual errors, MAE and RMSE are not the criterion.